MY STORY · LAHORE TO PORTSMOUTH AND BACK

I'm Asad Ikram. This is my story.

Data engineer • Chevening Scholar 2024/25 • Co-founder and CTO, Artemis AI

I came to computer science with no computing background and opened my Fiverr account in December 2018. That was the start of **nearly eight years** of data work: SQL tutoring first, then large-scale web and social media data collection, and the pipelines and models that turn that data into decisions people can trust. This page tells the story in order, from school in Lahore to the company I am building now.

Lahore school — Karachi 2017 — Chiniot-Faisalabad 2019 — Lahore 2022 — Portsmouth 2024 — Lahore 2025

442
completed Fiverr orders
FIVERR · FROM NOV 2019

30+
countries my Fiverr buyers came from
FIVERR BUYER MAP

300+
crawlers running daily at Dubizzle Labs
LAHORE · 2022 TO 2024

10
countries covered across six products
DUBIZZLE LABS

70,000+
applications in my Chevening cycle, for around 1,500 awards a year
CHEVENING 2024/25

255,446
public comments analysed in my MSc dissertation
PORTSMOUTH · 2025

UPDATED 25 OCTOBER 2026

THE TIMELINE

From Lahore classrooms to a PhD application

Twelve chapters, in order. Each one has the story, the key facts, and the tools I was using at the time.

School years · CHAPTER 01 · LAHORE

Growing up in Lahore

I grew up in Lahore. I did my O Levels at Crescent School and my FSc at Crescent College, both in the city. At school I was on the athletics team, running the 200, 400 and 800 metres and relays and competing in the long jump and high jump, and I played for the school football team. At college I took part in speech competitions and played for the college football team. Computers were not part of my schooling; I met them at university.

KEY FACTS

O Levels
Crescent School, Lahore

FSc
Crescent College, Lahore

Urdu and English
Urdu is my first language; English is my working one

O Levels · FSc · Urdu · English

2017 · CHAPTER 02 · KARACHI

Computer science, from zero

In August 2017 I started a BS in Computer Science at FAST-NUCES, a federally chartered university widely regarded as one of Pakistan's leading institutions for computer science. I began at its Karachi campus, far from home, with no computing background at all.

Good teachers and online courses changed that. I found I liked the subject, and then that I wanted to do more with it than the syllabus asked, because I knew I could.

KEY FACTS

Aug 2017
BS Computer Science starts at FAST-NUCES

Karachi campus
Aug 2017 to Jan 2019

Computer science · Karachi campus

2018 CHAPTER 03 · SUMMERS

Summers with SQL

I spent my university summers teaching myself Python and SQL, and I got properly good at SQL and databases. I took online and paid courses in SQL, data analytics and data warehousing on top of my degree.

On 20 December 2018 I opened a Fiverr seller account, offering SQL help and tutoring for students. Teaching is where the work started: through Fiverr I went on to teach data courses to fellow students and to international students, and client projects grew from there.

KEY FACTS

Dec 2018
Fiverr seller account opened

First role
SQL tutor for students

SQL · Python · Databases · Data warehousing · Teaching

2019 to 2020 CHAPTER 04 · CHINIOT-FAISALABAD

A new campus and the first orders

In January 2019 I transferred to the FAST-NUCES Chiniot-Faisalabad campus, where I finished my degree. In November 2019 my first Fiverr order arrived, and from 2020 work came in regularly.

I did not stay in one lane. I moved from tutoring into data analytics, web data collection, automation scripts, ETL and data pipelines, and I built back-end and front-end web projects with Django and Bootstrap. In June 2020 I delivered a trade-directory data project for a direct client in the UK.

Five-star reviews from clients and professionals abroad were a huge confidence boost. I knew I could make it.

KEY FACTS

Jan 2019
Moved to the Chiniot-Faisalabad campus

Nov 2019
First Fiverr order

200+
websites I collected data from as a freelancer

Python · Django · Bootstrap · ETL · Data pipelines · Automation · Web data collection

2021 CHAPTER 05 · FINAL YEAR

Final year, first remote teams

My final year was busy. From March to May 2021 I did a remote technology internship with an Indian company: data mining, data analytics, data acquisition and lead generation, and automation that cut manual work. Between June and September I added the Google Data Analytics Certificate, Python Data Structures from the University of Michigan on Coursera, and HackerRank's Python and Problem Solving certificates. In August I reached Level 2 on Fiverr for the first time.

For my final-year project, a team of three of us designed a **Custom Network Simulator**: a Python simulator of the Wi-Fi (IEEE 802.11) MAC layer with one-click dataset export and graphs, so researchers could generate training data for machine-learning models. Established tools such as NS2, NS3 and OMNeT++ are hard to extend with machine learning, and that was the gap. The idea was my supervisor's and it sat far from my data work, which is exactly why I took it on. It taught me how network protocols work and how web traffic is handled.

In November 2021 a Singapore-headquartered property-data company with a Japan office hired me for a short remote contract as a data engineer. It was my first taste of organised remote work, and of real responsibility. After about three months I left to focus on my final-year project.

KEY FACTS

Mar to May 2021
Remote technology internship, India

Aug 2021
Fiverr Level 2 reached for the first time

Grade A
Final-year project

Nov 2021
Short remote contract, Singapore company

`Python` `IEEE 802.11` `Qt` `MySQL` `Data analytics` `Lead generation`

COURSES ACROSS THE DEGREE

`Database Systems` `Design and Analysis of Algorithms` `Linear Algebra` `Calculus` `Probability and Statistics`

2022 CHAPTER 06 · DUBIZZLE LABS, LAHORE

A blank page at Dubizzle Labs

My degree was conferred on 31 January 2022. I had accepted an offer from Dubizzle Labs while my final exams were still running, and on 1 February I started in Lahore as an Associate Software Engineer. Dubizzle Labs is the technology arm of Dubizzle Group, whose brands at the time included dubizzle and Bayut in the Middle East and Zameen and OLX in Pakistan.

The company wanted competitor analysis from public web data, and there was no architecture for it yet. With one colleague in the Dubai team, I designed and built the first one on AWS: S3 as the data lake, Glue crawlers, Athena, and a Redshift warehouse in bronze, silver and gold layers, with dashboards in Periscope (Sisense).

This is where I moved from collecting data to using it. I used machine learning for the first time there, estimating competitor growth and revenue from public data, and studying what people discussed, how much volume each site carried, and what worked on each platform and in each region. That summer I took five DataCamp courses in data engineering and Python to go deeper.

KEY FACTS

1 Feb 2022
Associate Software Engineer, Lahore

First
competitor-analysis scraping architecture at the company, built with one Dubai colleague

4 languages
English, Arabic, Urdu and Hindi data

`AWS S3` `AWS Glue` `Athena` `Redshift` `Python` `Scrapy` `Selenium` `Periscope (Sisense)`

2023 CHAPTER 07 · RUNNING THE FLEET

300 crawlers, every day

In February 2023 I was promoted to Data Engineer. The architecture had become a fleet: more than 300 crawlers running daily, covering six products across ten countries (Pakistan plus nine Middle East and North Africa markets), with the data maintained together with the Dubai team. I ran it, monitored it, extended it and kept crawlers repaired on time. To make it more reliable I added unit tests, CloudWatch alarms and Retool QA apps, built ElasticSearch pipelines, and tuned fact tables for warehouse performance.

I also led the automation scripts that served marketing, finance, SEO and the CEO's team, led hiring for the scraping project, trained colleagues on ETL, and introduced juniors on my team to freelancing.

In May 2023, with Dubizzle's written consent, I started a second, part-time remote role as a data engineer (independent contractor) for **Fix.com** in Canada, an online retailer of appliance and lawn-equipment repair parts and a division of Eldis Group. My job there is data acquisition: finding target sites, building crawlers for retail and parts catalogues, and automating manual work. Flows and pipelines I built in those first months still run. Later I added OCR-based machine learning to read parts catalogues from PDFs and models to fill gaps in parts data.

In November 2023 I sent my first Chevening application. At its centre was an idea I still work on: a cross-platform system for analysing public social-media discussion responsibly, so public bodies can read public reaction and make evidence-based decisions at the right time. My example was the business shutdowns in Karachi in September 2023, after fuel and electricity price rises were announced close together. In the same application I named Dr Ella Haig's teaching in Intelligent Data and Text Analytics as a reason to choose Portsmouth, before I had ever met her.

KEY FACTS

Feb 2023
Promoted to Data Engineer

300+
crawlers running daily

6 products
across 10 countries

May 2023
Fix.com, Canada, part time

Nov 2023
First Chevening application

`CloudWatch` `Retool` `ElasticSearch` `Unit tests` `Scrapy` `Python`

2024 CHAPTER 08 · CHEVENING

Senior, selected, and on a plane

In February 2024 I became Senior Data Engineer: two promotions in two years. I led a team of three engineers, two in web scraping and one in data warehousing, and the data-warehouse team for Propforce, Zameen's real-estate affiliate sales platform. I did its data modelling from scratch with Redshift, dbt, Matillion and Airflow, using slowly changing dimensions and star and snowflake schemas.

The Chevening interview invitation came the same month, and I interviewed at the British Council in Lahore in March. I was conditionally selected in June, and my Final Award Letter arrived on 12 August 2024. I left Dubizzle that month, and in mid-September I moved to Portsmouth on my own to start an MSc in Data Analytics at the University of Portsmouth. New country, new city, new people. The Chevening orientation followed in November 2024, and through the University of Portsmouth's Student StartUp team I registered for Venturefest 2024.

CHEVENING SCHOLAR 2024/25

Chevening is the UK Government's international scholarship programme, funded by the FCDO and partner organisations, and it has run since 1983. My award came with a contribution from the University of Portsmouth.

I was selected on my first attempt, in a cycle of **more than 70,000 applications** worldwide (Chevening's figures imply about 72,000). Chevening awards around 1,500 scholarships a year to applicants from over 160 countries and territories; 46 went to Pakistan that year.

From September 2024 I also worked part time and remotely for **Prefe**, a Milan start-up that provides price, promotion and distribution intelligence for consumer-goods brands, as one of its early engineers. I moved its web data collection onto Scrapy and built pipelines into its databases. In December, with the work done, we parted on good terms.

That same December I won a part-time data engineering role at **M+C Saatchi Fluency**, the data consultancy of M+C Saatchi Group (a London-listed creative network with operations in about 23 countries), through a web-scraping challenge. I work remotely for the London team and daily with colleagues in New York.

KEY FACTS

Feb 2024
Senior Data Engineer, leading 3 engineers

12 Aug 2024
Chevening Final Award Letter

Sep 2024
MSc starts in Portsmouth; Prefe, Milan

Dec 2024
M+C Saatchi Fluency, London

`Redshift` `dbt` `Matillion` `Airflow` `BigQuery` `Snowflake` `Star schema` `Scrapy`

2025 CHAPTER 09 · PORTSMOUTH

A dissertation, a start-up, a Distinction

The MSc is where I got much more interested in machine learning and AI. I worked hands-on with Apache Spark on the university's computing cluster and used it in my projects. I made friends from Italy, France, Spain, Greece, Nigeria, the USA and India, and I mentored classmates who arrived without a data background.

In February 2025 I co-founded **Artemis AI** in the UK as CTO, with a partner I met while living there. In June and July I was also Lead Data Engineer, part time, at a European proptech start-up, leading three developers on real-time auction data pipelines and its Azure-to-AWS migration work. I left when that migration project stopped, to focus on my MSc and my other work.

My dissertation, supervised by Dr Ella Haig, was a cross-platform sentiment analysis of public reaction to UK economic policies: **255,446 public comments** from X/Twitter, Reddit, YouTube and Quora, collected through official APIs and licensed data access, under university ethics approval and with user IDs hashed. I compared a VADER lexicon baseline, a BiLSTM and a fine-tuned BERT model, calibrated them, tested differences with McNemar and bootstrap tests, forecast trends with Prophet, measured emotion with the NRC lexicon, and explained predictions with LIME and integrated gradients. It earned a mark of 78.

In September 2025 my study ended. I was welcomed into the Chevening alumni network, said goodbye at the Chevening Farewell at Alexandra Palace in London, and took ownership of Amazon's social-media analytics pipeline at Fluency. From late 2025 I also started working as a data engineer for CXG, a Dubai-based customer-experience consultancy, on data from global fashion and luxury brands. My MSc was awarded with **Distinction** on 8 December 2025, and that month I came home to Lahore.

KEY FACTS

Distinction
MSc Data Analytics, average 70.5

78
Dissertation mark

255,446
comments across 4 platforms

Feb 2025
Co-founded Artemis AI as CTO

Dec 2025
Back to Lahore

`Apache Spark` `PySpark` `BERT` `BiLSTM` `VADER` `Prophet` `LIME` `AWS Batch` `Step Functions` `Playwright`

Jan to Jul 2026 CHAPTER 10 · LAHORE

Back in Lahore, still building

Artemis AI Ltd was incorporated in the UK in January 2026. From Lahore I carried on with Fluency, Fix.com and Artemis, and went back to CXG for further engagements. One CXG problem I enjoy: the same product attribute is written differently by region and language, so "K18WG", "18K white gold" and "750/1000 white gold" have to become one shared vocabulary before anyone can compare brands.

By February I was also co-leading **ScrapeSync**, a small web-scraping team. In May I started **web-scraping-guide.com**, a free, no-signup engineering guide to web data collection, including its legal and ethical limits. It is how I wanted to give back after my Chevening year.

In June I became the sole designer and operator of Fix.com's crawler platform. **ScrapeYard** is a FastAPI control plane with a crawler registry, scheduling, live logs, on-demand data-quality checks, named logins with per-action audit trails, and Google Drive delivery. When a crawler breaks, AI can propose a fix, but it comes back as a diff that a person reviews; nothing commits on its own. It runs on an inexpensive VPS rather than managed cloud services.

In mid-2026 I also began advising a US travel-technology company (under NDA) on its large-scale data-collection architecture. And in July, with the Fluency team, I built a one-off collection across 20 powersports forums, plus Reddit through its official API.

KEY FACTS

Jan 2026
Artemis AI Ltd incorporated

May 2026
Free guide launched

10.03M
rows delivered by ScrapeYard across 82 datasets (Oct 2026)

~331
ScrapeYard commits, all mine, Jun to 4 Oct 2026

`FastAPI` `Scrapy` `Scrapyd` `APScheduler` `GitHub Actions` `Linux and systemd` `Google Drive API` `Reddit API`

Aug to Oct 2026 · CHAPTER 11 · AGENTS AND A PAPER

AI agents, a preprint and PhD applications

In August 2026 a developer advocate at Zyte got in touch because of the guide. Zyte's developer-relations team then invited me to review a pre-release version of its open-source agentic scraping toolkit, and later gave me two complimentary passes to the Web Data Extract Summit in Dublin.

At Fluency, **Crawler IQ**, the system I built and own for Ford, is a fleet of more than 60 car-owner forum crawlers with two AI agents: a healer that diagnoses a broken crawler, regenerates it and tests it on live pages, and a builder that creates new crawlers from a brief. Their changes ship only after verification. By 9 October 2026, across 1,697 runs, the healer had fixed 86% of broken crawlers and the builder had succeeded on 92% of builds.

I also own Amazon's CORE social-media analytics pipeline, whose master table holds about 2.2 million rows. In October I refactored it onto Fluency's shared data platform and checked parity file by file: all 138 exported files came out byte-identical. Alongside that, I built a 12-month US consumer discourse study for M+C Saatchi's New York team, with more than 100,000 consumer mentions, an 18-theme codebook that the team signed off, and a prompt version recorded on every machine label. I also support a UK government social-media analytics pipeline, and work on projects for global brands in consumer health, automotive, e-commerce and sport.

14 OCTOBER 2026 · FIRST PREPRINT

My first preprint went live on arXiv (not peer reviewed), written with Dr Ella Haig: a protocol for checking whether platforms change what machine-labelled sentiment means. More in Research and PhD.

The same month I applied to PhD programmes starting in 2027 in Europe, the UK and Singapore, with programmes in the US next on my list.

KEY FACTS

1,697 runs
86% healed, 92% of builds succeeded

~2.2M
rows in the Amazon CORE master table

138 of 138
files byte-identical after the refactor

arXiv:2610.16757
preprint, 14 Oct 2026

`Claude over MCP` `Scrapy` `FastAPI` `AWS Lambda` `ECS Fargate` `PostgreSQL` `Tableau` `BERTopic`

Next CHAPTER 12 · LATE 2026 AND 2027

What comes next

In November 2026 I plan to be at the Web Data Extract Summit in Dublin, which Zyte calls the world's largest web-scraping conference. Artemis AI opens a free invited beta on iOS and Android on 15 December 2026, with a public launch planned for 2027.

With Dr Haig I plan to submit our work to ICWSM, a CORE A-ranked AAAI venue for social-media research. Nothing is promised until reviewers say so. And if I am admitted, I start a PhD in 2027.

KEY FACTS

10 to 11 Nov 2026
Web Data Extract Summit, Dublin

15 Dec 2026
Artemis AI invited beta

2027
PhD start, if admitted

`Measurement` `Cross-platform NLP` `Multilingual text` `Product launch`

FREELANCE, SINCE 2018

Clients in more than 30 countries

Fiverr is where I learned to work with clients anywhere. **442 completed orders**, more than 300 five-star reviews and Level 2 Seller status (first reached August 2021), with buyers spread across these countries.

442	300+	33	200+
COMPLETED ORDERS	FIVE-STAR REVIEWS	COUNTRIES LISTED HERE	WEBSITES COLLECTED FROM

From a student room in Pakistan to buyers on six continents

Every arc starts in Lahore. Fiverr's own analytics put my buyer map at 22 percent of the world: North America, Western Europe and the Nordics, the Gulf and North Africa, South and Southeast Asia, Australia, South Africa and Peru.

Most of those orders were data work: SQL, analytics, and collecting and cleaning web data for people who needed it to make a decision.

LAHORE · BUYER COUNTRY · ORDERS

Americas — 3 COUNTRIES

United States · Canada · Peru

Western Europe — 7 COUNTRIES

United Kingdom · Ireland · France · Belgium · Netherlands · Germany · Switzerland

Nordics — 4 COUNTRIES

Denmark · Norway · Sweden · Finland

Southern and Eastern Europe — 8 COUNTRIES

Spain · Portugal · Italy · Greece · Poland · Romania · Ukraine · Russia

Middle East and North Africa — 4 COUNTRIES

Türkiye · Saudi Arabia · Egypt · Algeria

South Asia

2 COUNTRIES

Pakistan · India

Southeast Asia and Oceania

4 COUNTRIES

Thailand · Malaysia · Philippines · Australia

Sub-Saharan Africa

1 COUNTRY

South Africa

Read from Fiverr's buyer-geography map, so treat the list as approximate; a few countries that are hard to read on the map are left off. Fiverr's map labelled my reach **"22% world domination"**. Outside Fiverr I also worked directly for clients in the UK (2020) and California (2022 and 2026).

THE THREAD

Every system I built left me with the same question: do these numbers really mean what we think they mean? Industry gave me the question. Research is where I want to answer it, and then bring the answer back to the people who rely on it.

LAHORE · PORTSMOUTH · LAHORE

COMMUNITY AND MENTORING

Helping other people get started

Freelancing showed me what I could do. The most useful thing I can do with that is help other people get there faster.

UNIVERSITY · PUNJAB

First in their families

At university I helped several fellow students from remote parts of Punjab start freelancing. They were the first in their families to reach a university like ours, and money was tight. I set up their freelance accounts, passed clients to them, taught them the skills, and kept mentoring them outside work.

FIVERR · TEACHING

Data courses for students

Through Fiverr I taught data courses to fellow students and to international students, starting with the SQL that got me going.

DUBIZZLE LABS

Juniors on my team

I mentored the juniors on my team, trained colleagues on ETL, and introduced juniors to freelancing.

PORTSMOUTH

Classmates new to data

On my MSc I mentored classmates who arrived without a data background.

SCRAPESYNC

A small team I co-lead

ScrapeSync is a small web-scraping services team and brand that I co-lead. It lets me take on bigger freelance work and lead a team doing it.

WEB-SCRAPING-GUIDE.COM · SINCE MAY 2026

A free field guide to web data collection

Asad's Guide to Web Scraping 2026 is a free, no-signup engineering guide to resilient, large-scale web data collection that respects platform terms, using official APIs where available and polite rate limits. It covers the legal and ethical limits too, and I update it weekly.

23	255	91	11
SECTIONS	CARDS	LIBRARIES COMPARED	PIPELINE DESIGNS

Figures as of 30 September 2026. Readers have written in, a recruiter among them, and it is how Zyte's developer-relations team found me.

ZYTE · DUBLIN

Invited to review

In August 2026 Zyte's developer-relations team invited me to review a pre-release version of its open-source agentic scraping toolkit. They also gave me two complimentary passes to the Web Data Extract Summit 2026 in Dublin.

A PROMISE FROM 2023

What I told Chevening I would do

In my 2023 Chevening application I wrote that I wanted to run webinars and data-collection courses at universities, help under-privileged students into freelancing, and build a bridge between Chevening scholars and local talent in Pakistan. Some of that is still a plan. The free guide and the mentoring are the parts I have started.

RESEARCH AND PHD

One question since 2023

Brands, start-ups and public bodies act on numbers taken from social media. I want to know when those numbers can be trusted, across platforms and across languages.

2023

Chevening essay

A cross-platform system to read public reaction responsibly and inform decisions.

2025

MSc dissertation

255,446 comments on UK economic policy, four platforms. Distinction.

2026

Preprint

Is the platform part of the measurement? With Dr Ella Haig.

2027

PhD, if admitted

Does machine-labelled opinion mean the same across platforms, languages and scripts?

MSC DISSERTATION · 2025

What my dissertation taught me

"Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies", supervised by Dr Ella Haig. To check my labels I hand-labelled 200 posts myself; agreement with VADER was a kappa of about 0.78. The bigger lessons were about what I had got wrong:

01 The classifier's apparent accuracy measured agreement with the lexicon that made its labels, not accuracy.

02 I treated the platforms as if they measured opinion the same way. They do not have to.

03 The English-only pipeline dropped posts written in Urdu and Hindi.

Together, those three lessons became the preprint and my PhD direction.

PREPRINT · NOT PEER REVIEWED

Is the Platform Part of the Measurement?

> Ikram, A. and Haig, E. (2026). "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." arXiv:2610.16757. Version 1.1, 25 pages; cs.LG, cross-listed in cs.CL, cs.CY and cs.SI.

It treats each platform like a survey mode and sets out a six-step protocol:

1. Fixed anchor events
2. A generalizability study against human labels
3. Measurement invariance and DIF
4. A multilevel model
5. Score linking
6. A 500-replication known-truth simulation

The simulation shows that pooling sentiment across platforms misleads when platforms measure differently and the mix of platforms shifts, and that relative linking fixes it. The strongest headline comparison holds only when the true measurement functions are assumed known; with a realistic reference point the advantage shrinks. It has not yet been applied to real data. An experienced reviewer judged the idea sound but found the hand-labelled set and corpus from my MSc, which motivate it, small. That tells me what to fix next.

Next: ICWSM is a CORE A-ranked AAAI venue for social-media research; a submission with Dr Haig is planned.

arxiv.org/abs/2610.16757 github.com/asadfix/platform-measurement

DOI 10.5281/zenodo.23105129

PHD DIRECTION · APPLYING FOR 2027

Where I want to take it

My PhD proposals ask whether opinion labelled by language models and classifiers means the same thing across platforms, languages and scripts. The plan uses designs that hold the content fixed, and a Roman Urdu political-stance gold set.

The motive is practical. I want to solve the problems I meet at Artemis AI academically first, and then in the product. I am applying to programmes in social-media research and web data collection, starting in 2027. Several of the professors I wrote to replied positively, and one gave critical feedback that sharpened the proposal.

LANGUAGES AND SCRIPTS IN SCOPE

URDU SCRIPT	LATIN SCRIPT	LANGUAGE	CODE-MIXED
اردو Urdu	Roman Urdu	English	Hinglish

WHAT I AM BUILDING

Artemis AI

Large companies can afford social-media intelligence. Most creators and small businesses cannot. Artemis AI is a UK start-up that aims to give them the same kind of insight, so they can compete. I am its co-founder and CTO.

SINCE FEBRUARY 2025 · UK

Social-media intelligence for Facebook and Instagram pages

Artemis connects to a page through Meta's official Graph APIs with read-only access, reads its posts, comments and insights, and turns them into advice the owner can act on. It never posts or replies.

- Pre-launch · free invited beta on iOS and Android from 15 Dec 2026 · public launch planned for 2027

E

Meet Art-E

Art-E, our AI advisor, tells a page owner what their audience is saying and what to do next.

- What to reply to
- When to post
- Whether a draft will land
- Early warning of comment storms
- Possible collaboration partners

48	~10	3.39M	4
models in the architecture, across 7 families	running in weekly production	public comments collected from Facebook pages	contractors I lead: AI/ML, back end, product design, legal

HOW IT IS BUILT, LAYER BY LAYER

01 **Sources**
Facebook and Instagram Graph APIs, read-only.

02 **Collection**
Posts, comments and insights; hourly growth curves for posts up to 14 days old; a 90-day backfill when a page connects.

03 **Warehouse**
Apache Iceberg tables with an Athena star schema. Facebook and Instagram sit side by side and are never added together.

04 **Models**
Our own NLP models first, a large language model only below a confidence floor. Five DeBERTa-v3 comment models read each comment with its post; images go through OCR and CLIP; video through frames, speech-to-text and OCR.

05 **Serving**
PostgreSQL with pgvector, and a model registry.

06 **Art-E**
Retrieval over the page's own data with a safety guard. Every number must come from a checked tool, and unsourced numbers are blocked. An LLM judge runs on 200 test questions, and the next-move picker holds back 10% to prove it helps.

07

App

iPhone and Android, hosted in AWS London.

RESPONSIBLE BY DESIGN

- Read-only Meta permissions
- Commenter IDs are never stored, and tagged names become "user"
- Audience overlap kept only as page-pair counts of 5 or more
- A UK-only data rule
- A legal check that no AI vendor trains on customer content
- Comment storms judged against each page's own normal

MY PART, AND THE TEAM'S

As CTO I lead the architecture, evaluation and QA dashboards, data governance and permissions, crisis definitions, and the planning and tracker. I set up the mobile app repository with CI and a protected main branch, and I take part in pitching.

I brought four contractors into the team, and our AI/ML and back-end engineers wrote most of the model pipelines. The whole company team is about eight people, including the co-founders.

WHAT IT TAUGHT ME

Agreement between models is not accuracy, so our tracker calls it an accuracy proxy. Banter, dialect and non-English text are where models fail first, and our OCR reads only English. Those lessons are a big part of why I want a PhD.

THE THREAD THROUGH ALL OF IT

Numbers from social media can be quietly wrong

LANGUAGE

Posts in other languages get dropped.

TONE

Banter gets read as anger.

ACCURACY

Agreement between models gets passed off as accuracy.

PLATFORMS

Pooled platforms shift results when the platform mix changes.

MY ANSWER HAS TWO HALVES

Measure it, then build the checks in

MEASUREMENT

The protocol in my preprint, and the PhD that should follow it.

ENGINEERING

Verification gates on AI-repaired crawlers, versioned prompts, codebooks that people sign off, clearly flagged estimates, read-only data access and no stored commenter identities.

My personal goal, unchanged since my Chevening application, is to help public bodies read public reaction responsibly and decide at the right time. Today I pursue it through work like the UK government pipeline I support at Fluency.

"My journey has not stopped; it has started."

ASAD IKRAM, OCTOBER 2026

Thanks for reading

If any of this overlaps with what you are working on, I would like to hear from you.

EMAIL

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LINKEDIN

/in/asad-ikram98

PORTFOLIO

asad-ikram-mc.github.io/portfolio

Lahore, Pakistan · Remote worldwide

Story updated 25 October 2026